Age: The disease may occur at any age but about 80 % of the cases occur after the age of 45.

Sex: In the younger age groups, diabetes is more commonly seen in males than in females. In middle age, women are more affected and the chances increase with pregnancy and increasing parity.

Obesity: There is a strong association between diabetes and obesity. In obesity, there is hyperinsulinaemia and insulin resistance or impaired insulin uptake by receptors in target tissues. Obese people are also less physically active than normal weight individuals which increase the risk of diabetes.



Why weight is a risk factor for Diabetes?

- Overweight people have fewer available insulin receptors.
- More fat requires more insulin, promoting insulin resistance.
- Fat cells release free fatty acids which interfere with glucose metabolism.

Dietary factors/Nutritional factors: A high intake of sugar has also been related to obesity and may predispose to diabetes. A low intake of fibre due to consumption of refined foods is associated with high prevalence of Diabetes.

Infections: Type I diabetes usually occurs after viral infection, which brings about an auto immune reaction that may destroy the beta cells of the pancreas and impairs insulin secretion.

Stress: It elucidates adrenaline and thereby precipitates diabetes.

Secondary Diabetes

Diabetes can also occur secondary to the following disorders:

- ❖ Diseases which destroy the pancreas and lead to impaired secretion and release of insulin. E.g., pancreatitis, haemochromatosis, carcinoma of pancreas and pancreatectomy.
- ❖ Abnormal concentration of certain hormones in the circulation which are insulin antagonist. These include:
  - Growth hormone
  - Adrenocortical hormone (Cushing syndrome, Addison's disease, Hypopituitarism)
  - Adrenaline
  - Thyroid hormone



Activity : 1

Meet any two diabetics and find out the reason for them developing diabetes.

1. Diabetic 1 -
2. Diabetic 2 -



Women of reproductive age who have developed polycystic ovarian syndrome (PCOS) are at an increased risk of type 2 diabetes. PCOS is a hormonal disorder characterized by enlarged ovaries containing fluid filled cysts. Women with PCOS have irregular menstrual cycles and high circulating levels of male hormones like testosterone. Insulin resistance and impaired glucose tolerance are manifestations of PCOS.



Activity : 2

Sita is overweight, but none in her family has diabetes. Should she be really concerned with her weight?

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